CMSC 104 Section 2
Spring 2026
Sample Quiz 3

Instructions:

This is a Sample Quiz for Quiz 3. It contains the same number and type of questions that students will encounter on the actual quiz.

This quiz is to be completed on paper, using a pen or pencil. Please make sure you write your name on the paper. When you are finished you may turn in your paper and leave the room.

This quiz is “open-book, open-notes.” You MAY use any of the following: your laptop or phone, a book; your notes; the instructor’s notes and code from the GitHub site; an IDE such as IDLE or PyCharm (if you wish to test your code to make sure your answer is correct). You MAY NOT use: any other person, whether in this classroom or not (e.g., you may not text your friend the computer science major), or any AI-support source such as ChatGPT, Gemini, CoPilot, Claude Code, etc. If you have a question about this, please raise your hand and ask.

Section 1: True/False and Multiple Choice (40 points)

There are 10 questions in this section. Each is worth 4 points. No partial credit is given for any question in this section. If you change your answer during the quiz, please make sure you clearly indicate which is your final answer.

1. True or False: a “for i” loop is the most general type of loop in Python. Any iteration problem that can be solved in Python, can be solved with a “for i” loop.
    a. True
    b. False

2. What is the sentinel in a sentinel while loop?
    a. The value that when encountered causes execution of the loop to stop
    b. The break statement that causes execution of the loop to stop
    c. The first assignment to a variable in the while loop’s boolean expression
    d. None of the above is correct

3. Suppose you have a string variable, `s = "Department of Electrical Engineering and Computer Science"`. What does the statement `print (s[:9])` print?
    a. `Department of Electrical Engineering and Computer Science`
    b. `Department`
    c. `Departmen`
    d. `Nothing, it returns an error`

4. True or False: lists and dictionaries are immutable variables in Python. This means that you can change their values in place, without having to reassign the variable and get a new location in memory.
    a. True
    b. False

5. How do you create a new blank two-dimensional list, `my_list`, in Python?
    a. `my_list =[]`
    b. `my_list = {}`
    c. `my_list = ()`
    d. `my_list = [ ][ ]`

6. What is the best way to develop software?
    a. Write a little, test a little - do it in small chunks and make sure each chunk works before moving on
    b. Write it all at once and hope it all somehow magically works when you're done.
    c. Are you still reading? The answer is a.
    d. Seriously. Just pick a.

7. Suppose that you have a list my_list in Python. If you wanted to get rid of the ninth element in the list, without regard to its value, which method would you use?
    a. `append()`
    b. `pop()`
    c. `remove()`
    d. `Any or all of the above could be used.`

8. When you are referencing an element in a two-dimensional Python list, which element comes first, the row or the column
    a. Row
    b. Column

For questions 9 and 10, assume we have the following dictionary of people who teach CMSC 104:

```
profs_104 = {
```

```
    "zak":1,
    "arsenault":2
}
```

9. What value does n have when we execute the statement
   `n = profs_104.pop("Joshi")`

   a. There's no value; the program crashes due to an error
   b. None
   c. -1
   d. None of the above answers is correct

10. What value would n have if we instead executed the statement
    `n = profs_104["Joshi"]`

    a. There's no value; the program crashes due to an error
    b. None
    c. -1
    d. None of the above answers is correct

Section 2: Short answer (40 points)

There are 8 questions in this section. Each is worth 5 points. Partial credit IS given for questions in this section, so please make an attempt to answer each question.
Note: “short answer” means one or two sentences. It should not take you more than that to answer a question. Continuing to write paragraph after paragraph is probably not going to help make your answer more correct.

11. Give an example of a while loop that would run zero times, on purpose.

    When you ask the user for input, and then check to see if the input is valid. If the input is invalid, your loop runs to let the user know the error and ask for input again. If the user input is valid the first time, your loop will never run

12. Write Python code that creates a two dimensional list of the 49 integers starting at 1 and ending at 49. Your list should have seven rows of seven columns each.

```
my_list = []
for i in range(1, 8):
   new_row = []
   for j in range(1, 8):
       new_row.append((i-1)*7 + j)
   my_list.append(new_row)
```

13. When everybody’s healthy, here’s the Baltimore Orioles’ starting infield: first base, Ryan Mountcastle; second base, Jackson Holliday; shortstop, Gunnar Henderson; third base, Coby Mayo. Create a dictionary, birds_infield, containing this data.

```
birds_infield = {
   "first": "Ryan Mountcastle",
   "second": "Jackson Holliday",
   "short": "Gunnar Henderson",
   "third": "Coby Mayo"

}
```

14. You must close a data file in Python when you are finished using it. If you don't, the file system could become unstable. Using the "with open" statement as we do, how do you close the file?

    "With open" automatically closes the file when you're done, so you don't have to do anything to close it.

15. When adding items to a list, what's the difference between the `append()` method and the `insert()` method?

append() puts the new element at the end of the list. insert() puts it wherever in the list you want it.

16. List two reasons a while loop might be an infinite loop.

a. You never change the variable in your boolean expression, so that the value never changes
b. You change the variable in your boolean expression, but your logic is wrong and the boolean expression is never false.

17. Suppose that I have the following for i loop:

```
for i in range(1, 10):
      print( i, " squared is ", i**2)
```

Rewrite this as a while loop.

```
'i = 1
while i < 10:
   print( i, " squared is ", i**2)
   i = i + 1
```

18. What must be true about the keys in a dictionary, in terms of types and uniqueness?

    They must be immutable and unique.

Section 3: Programming (20 points)

There are two questions in this section; each is worth 10 points. Partial credit IS given for questions in this section.
You do not need to write comments with your code, unless you think that it will help the grader more clearly understand exactly what you're trying to accomplish.

19. Write a program that:

- Sets the value of pi to be 3.14159 (just use an assignment statement; there is no need to import the math library)
- Then uses a while loop to print the first ten multiples of pi. That is,
    - Print "1 * pi = 3.14159
    - Then print 2* pi = 6.28318
    - Then prints 3*pi, 4*pi, and so on; up through 10*pi
- You MUST use a while loop in this program! No "for" loops permitted.

```
pi = 3.14159
i = 1
while i <= 10:
   print( i, " times pi is ", i*pi)
   i = i + 1
```

20. Write a program that creates a dictionary of students and their majors and grade-point averages. .
    The main program must first create an empty dictionary, and send that as an argument to a function called get_data.
    The function get_data has this one parameter, which is a dictionary.
    The function must ask the user to input the student's name (a string); the student's major (a string) ; and the student's gpa (a float)
    The function must create a dictionary with these three key/value pairs.
    The function must then return the dictionary in a return statement. (Yes, we know you can make it work without a return statement; use one anyway.)
    Then the main program will print out the dictionary.

You must write the entire program, both main program and function.

```
def get_data(d):
   name = input("please enter the student name")
   d ["name"] = name
   major = input("please enter the student's major")
   d ["major"] = major
   gpa = float(input("please enter the student's GPA"))
   d ["gpa"] = gpa
   return d


if __name__ == "__main__":
   student = {}
   this_student = get_data(student)
   print(this_student)
```